

GCSE Mathematics - Foundation Tier

Paper 2 (Calculator) Practice - Set 3

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use $\pi = 3.142$.

1.

Round 3.467 to 1 decimal place.

(Total for Question 1 is 1 marks)

2.

Find the highest common factor (HCF) of 24 and 60.

(Total for Question 2 is 2 marks)

3.

- (a) Write 7800 in standard form. (1)
- (b) Write 0.0042 in standard form. (1)

(Total for Question 3 is 2 marks)

4.

Write 65% as a fraction in its simplest form.

(Total for Question 4 is 2 marks)

5.

A two-way table shows the colour preferences of 80 people for a new car.

	Red	Blue	Black	Total
Men	12	___	14	40
Women	8	18	___	40
Total	20	___	___	80

- (a) Complete the two-way table. (2)
- (b) Write down the number of women who preferred a black car. (1)

(Total for Question 5 is 3 marks)

6.

Substitute $x = -3$ into the formula $y = 2x + 5$.

Work out the value of y .

(Total for Question 6 is 2 marks)

7.

A train travels at an average speed of 80 mph and covers a distance of 220 miles.

Work out the time taken. Give your answer in hours and minutes.

(Total for Question 7 is 3 marks)

8.

The weekly wages, in pounds, of 7 part-time staff at a cafe are:

180 210 195 220 180 240 215

Work out the mean weekly wage. Give your answer to the nearest penny.

(Total for Question 8 is 3 marks)

9.

Two children, Riya and Sam, share some marbles in the ratio 5 : 3.

Riya receives 80 marbles.

Work out the total number of marbles they share.

(Total for Question 9 is 3 marks)

10.

Solve $(2x / 3) + 1 = 5$.

You must show all your working.

(Total for Question 10 is 3 marks)

11.

A right-angled triangle has a hypotenuse of 13 cm and one shorter side of 5 cm.

[Diagram: right-angled triangle with hypotenuse 13 cm and one shorter side 5 cm; the other shorter side is labelled x cm.]

Work out the length of the other shorter side x .

(Total for Question 11 is 3 marks)

12.

The pie chart shows the favourite ice-cream flavour of 240 children.

[Diagram: pie chart with sectors - Vanilla: 120 deg, Chocolate: 105 deg, Strawberry: 75 deg, Mint: 60 deg.]

- (a) Work out the number of children who chose Chocolate. (2)
- (b) A child is selected at random. Work out the probability that they chose Strawberry. (2)

(Total for Question 12 is 4 marks)

13.

The formula for the perimeter P of a rectangle of length L and width W is

$$P = 2L + 2W$$

- (a) Work out P when $L = 14$ cm and $W = 9$ cm. (2)

The formula for converting kilograms (k) to pounds (p) is approximately

$$p = 2.2 \times k$$

- (b) Work out p when $k = 35$ kg. (2)

(Total for Question 13 is 4 marks)

14.

After a 20% discount, a pair of trainers costs £64.

- (a) Work out the original price of the trainers. (3)

A different pair of trainers had an original price of £75. They were sold for £63.

- (b) Work out the percentage discount on these trainers. (2)

(Total for Question 14 is 5 marks)

15.

The cost of a meal is £42 before VAT.

VAT is added at a rate of 20%.

Work out the total cost of the meal including VAT.

(Total for Question 15 is 4 marks)

16.

A purple paint is made by mixing red and blue paint in the ratio 2 : 3.

- (a) Work out the amount of red paint needed to make 25 litres of purple paint. (2)

A painter has 18 litres of red paint and 24 litres of blue paint.

- (b) Work out the maximum amount of purple paint the painter can make. You must show your working. (3)

(Total for Question 16 is 5 marks)

17.

The table shows the engine size (in litres) and the CO₂ emissions (g/km) of 8 cars.

Engine	1.0	1.2	1.4	1.6	1.8	2.0	2.5	3.0
CO ₂	105	118	125	140	155	170	205	240

- (a) On the grid provided, plot these data as a scatter graph. (2)
- (b) Describe the correlation between engine size and CO₂ emissions. (1)
- (c) Draw a line of best fit. (1)
- (d) Use your line of best fit to estimate the CO₂ emissions of a car with an engine size of 2.2 litres. (1)

(Total for Question 17 is 5 marks)

18.

A closed cuboid has dimensions 6 cm by 8 cm by 10 cm.

[Diagram: closed cuboid 6 cm by 8 cm by 10 cm.]

- (a) Work out the volume of the cuboid. (2)
- (b) Work out the surface area of the cuboid. (3)

(Total for Question 18 is 5 marks)

19.

A survey of 60 people asked which streaming service they use most.

Service	Frequency
Netflix	20
Disney+	15
Prime	10
Apple TV	9
Other	6

Draw an accurate pie chart to represent this information. Label each sector clearly.

(Total for Question 19 is 5 marks)

20.

Three friends, Alex, Ben and Chris, raised money for charity.

Alex raised £ x . Ben raised £15 more than Alex. Chris raised twice as much as Alex.

Together they raised £155.

- (a) Show that $4x + 15 = 155$. (2)
- (b) Solve to find the value of x . (2)
- (c) How much did Chris raise? (1)

(Total for Question 20 is 5 marks)

21.

Priya saves money each month from her salary.

In January she saved £120.

Each month her savings increase by 5% on the previous month.

- (a) Work out how much she saves in March. (3)

(b) Work out the total amount she has saved by the end of March.

(3)

(Total for Question 21 is 6 marks)

22.

A bag contains 12 counters: 5 red, 4 blue and 3 yellow. A counter is taken at random.

(a) Write down the probability that the counter is yellow.

(1)

(b) Write down the probability that the counter is NOT yellow.

(1)

The counter is replaced and a counter is taken again. This is done 60 times.

(c) Work out the expected number of times a blue counter is drawn.

(3)

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER: 80 MARKS

Mark Scheme

Foundation Tier - Paper 2 (Calculator) - Set 3 | 80 marks

Question 1

- 3.5 - B1

Total: 1 marks

Question 2

- Factor lists or prime factorisation seen - M1
- 12 - A1

Total: 2 marks

Question 3

- (a) 7.8×10^3 - B1
- (b) 4.2×10^{-3} - B1

Total: 2 marks

Question 4

- 65/100 written - M1
- 13/20 - A1

Total: 2 marks

Question 5

- (a) Men Blue = $40 - 12 - 14 = 14$ - M1
- (a) Black total = $14 + (\text{Women Black})$; Blue total = $14 + 18 = 32$; Women Black = $80 - 32 - 20 = 28$;
OR Women Black = $40 - 8 - 18 = 14$ - A1
- (b) 14 - B1

Total: 3 marks

Question 6

- $2 \times (-3) = -6$ - M1
- $y = -6 + 5 = -1$ - A1

Total: 2 marks

Question 7

- $220 / 80 (= 2.75 \text{ hours})$ - M1
- $0.75 \text{ hours} \times 60 = 45 \text{ minutes}$ - M1
- 2 hours 45 minutes - A1

Total: 3 marks

Question 8

- Sum = $180 + 210 + 195 + 220 + 180 + 240 + 215 = 1440$ - M1
- $1440 / 7$ - M1
- £205.71 - A1

Total: 3 marks

Question 9

- One part = $80 / 5 = 16$ - M1
- Sam = $3 \times 16 = 48$ - M1
- Total = $80 + 48 = 128$ (marbles) - A1

Total: 3 marks

Question 10

- Subtract 1: $2x/3 = 4$ - M1
- Multiply by 3: $2x = 12$ - M1
- $x = 6$ - A1

Total: 3 marks

Question 11

- Pythagoras: $13^2 - 5^2$ (or $x^2 = 169 - 25$) - M1
- $169 - 25 = 144$ - M1
- $x = 12$ (cm) - A1

Total: 3 marks

Question 12

- (a) $105 / 360 \times 240$ - M1
- (a) 70 (children) - A1
- (b) $75 / 360$ seen - M1
- (b) $5/24$ (accept $75/360$ simplified) - A1

Total: 4 marks

Question 13

- (a) $2 \times 14 + 2 \times 9$ - M1
- (a) $P = 46$ (cm) - A1
- (b) 2.2×35 - M1
- (b) $p = 77$ - A1

Total: 4 marks

Question 14

- (a) After 20% discount, price = $0.8 \times$ original - M1
- (a) $64 / 0.8$ - M1
- (a) £80 - A1
- (b) $75 - 63 = 12$ (discount) seen - M1
- (b) $12/75 \times 100 = 16$ (%) - A1

Total: 5 marks

Question 15

- VAT = 0.20×42 or multiplier 1.2 seen - M1
- VAT amount £8.40 or 1.2×42 attempted - M1
- $42 + 8.40$ (or 42×1.2) - M1

- £50.40 - A1

Total: 4 marks

Question 16

- (a) $25 / 5 = 5$ (one part) - M1
- (a) Red = $2 \times 5 = 10$ (litres) - A1
- (b) Per litre of purple needs $2/5$ L red and $3/5$ L blue - M1
- (b) Red limit $18 / (2/5) = 45$ L; Blue limit $24 / (3/5) = 40$ L - M1
- (b) Minimum (40 L) is the maximum purple - A1

Total: 5 marks

Question 17

- (a) All 8 points plotted correctly - B2 (B1 if 6 or more correct)
- (b) Positive correlation (strong / very strong) - B1
- (c) Sensible line of best fit drawn - B1
- (d) CO₂ estimated at engine 2.2 L (accept 180 - 195) - B1

Total: 5 marks

Question 18

- (a) $6 \times 8 \times 10$ - M1
- (a) 480 (cm³) - A1
- (b) Three pairs of faces: $2(6 \times 8) + 2(6 \times 10) + 2(8 \times 10)$ - M1
- (b) $96 + 120 + 160$ - M1
- (b) 376 (cm²) - A1

Total: 5 marks

Question 19

- $360 / 60 = 6$ (deg per person) - M1
- At least 3 angles correct: Netflix 120, Disney+ 90, Prime 60, Apple 54, Other 36 (deg) - M1
- All 5 angles correct - A1
- Pie chart drawn within 2 deg tolerance - B1
- All sectors labelled - B1

Total: 5 marks

Question 20

- (a) Ben = $x + 15$, Chris = $2x$ identified - M1
- (a) $x + (x+15) + 2x = 4x + 15 = 155$ shown - C1
- (b) $4x = 140$ - M1
- (b) $x = 35$ - A1
- (c) Chris = $2 \times 35 = £70$ - B1

Total: 5 marks

Question 21

- (a) Multiplier 1.05 seen - M1
- (a) Feb $120 \times 1.05 = 126$ - M1
- (a) March $126 \times 1.05 = £132.30$ - A1

- (b) $120 + 126 + 132.30$ - M1
- (b) Addition completed - M1
- (b) £378.30 - A1

Total: 6 marks

Question 22

- (a) $\frac{3}{12}$ (or $\frac{1}{4}$) - B1
- (b) $\frac{9}{12}$ (or $\frac{3}{4}$ or 0.75) - B1
- (c) Expected = probability $\times$ trials - M1
- (c) $\frac{4}{12} \times 60$ - M1
- (c) 20 - A1

Total: 5 marks

Worked Solutions

Foundation Tier - Paper 2 (Calculator) - Set 3 | Detailed explanations

Question 1

The 1st decimal place is the tenths column. The digit there is 4. Look at the next digit (hundredths) to decide whether to round up: 6 ($\geq$ 5), so round up.

3.467 $\rightarrow$ 3.5.

Question 2

Prime factorisation:

$$24 = 2 \times 2 \times 2 \times 3.$$

$$60 = 2 \times 2 \times 3 \times 5.$$

Common prime factors: $2 \times 2 \times 3 = 12$.

HCF = 12.

Question 3

Standard form: $A \times 10^n$ where $1 \leq A < 10$.

(a) $7800 = 7.8 \times 1000 = 7.8 \times 10^3$.

(b) $0.0042 = 4.2 / 1000 = 4.2 \times 10^{-3}$.

Question 4

$$65\% = 65 / 100.$$

Find HCF of 65 and 100. Factors of 65: 1, 5, 13, 65. Factors of 100: 1, 2, 4, 5, 10, 20, 25, 50, 100.

HCF = 5.

$$65/100 = 13/20.$$

Question 5

(a) Men row: Red 12, Black 14, so Blue = $40 - 12 - 14 = 14$.

Women row: Red 8, Blue 18, so Black = $40 - 8 - 18 = 14$.

Column totals: Blue = $14 + 18 = 32$, Black = $14 + 14 = 28$. Check overall: $20 + 32 + 28 = 80$.

(b) 14 women preferred a black car.

Question 6

Substitute $x = -3$ into $y = 2x + 5$.

$$y = 2(-3) + 5 = -6 + 5 = -1.$$

Question 7

Time = distance / speed = $220 / 80 = 2.75$ hours.

0.75 hours = $0.75 \times 60 = 45$ minutes.

Total time = 2 hours 45 minutes.

Question 8

Sum of wages = $180 + 210 + 195 + 220 + 180 + 240 + 215 = 1440$.

Mean = total / number of staff = $1440 / 7 = 205.7142\dots$

To the nearest penny: £205.71.

Question 9

Riya gets 5 parts and these total 80 marbles, so 1 part = $80 / 5 = 16$ marbles.

Sam gets 3 parts = $3 \times 16 = 48$ marbles.

Total = $80 + 48 = 128$ marbles.

Question 10

$(2x / 3) + 1 = 5$.

Subtract 1 from both sides: $2x/3 = 4$.

Multiply both sides by 3: $2x = 12$.

Divide by 2: $x = 6$.

Check: $(2 \times 6)/3 + 1 = 12/3 + 1 = 4 + 1 = 5$.

Question 11

Pythagoras for finding a shorter side: $\text{shorter}^2 = \text{hypotenuse}^2 - \text{other}^2$.

$x^2 = 13^2 - 5^2 = 169 - 25 = 144$.

$x = \sqrt{144} = 12$.

Answer: $x = 12$ cm.

Question 12

(a) Chocolate sector = 105 deg out of 360 deg.

Number = $(105 / 360) \times 240 = 70$ children.

(b) Strawberry sector = 75 deg.

$P(\text{Strawberry}) = 75 / 360 = 15/72 = 5/24$.

Question 13

(a) $P = 2L + 2W = 2 \times 14 + 2 \times 9 = 28 + 18 = 46$ cm.

(b) $p = 2.2 \times k = 2.2 \times 35 = 77$ (pounds).

Question 14

(a) After 20% off, the sale price is 80% of the original.

$0.8 \times \text{original} = 64$, so $\text{original} = 64 / 0.8 = £80$.

(b) Discount in pounds = $75 - 63 = £12$.
Percentage discount = $(12 / 75) \times 100 = 16\%$.

Question 15

VAT at 20% on £42 = $0.2 \times 42 = £8.40$.
Total = $42 + 8.40 = £50.40$.
Or directly: $1.2 \times 42 = £50.40$.

Question 16

(a) Total ratio parts $2 + 3 = 5$. So 25 L of purple has $25 / 5 = 5$ L per part.
Red paint needed = 2 parts = $2 \times 5 = 10$ litres.
(b) For every 5 L of purple, you need 2 L red and 3 L blue. So 1 L of purple needs $2/5$ L red and $3/5$ L blue.
With 18 L red: $18 / (2/5) = 18 \times 5/2 = 45$ L of purple.
With 24 L blue: $24 / (3/5) = 24 \times 5/3 = 40$ L of purple.
The smaller limit is 40 L (blue limits the mix). Maximum purple = 40 L.

Question 17

(a) Plot 8 points.
(b) Positive correlation - as engine size increases, CO₂ emissions increase.
(c) Draw a straight line through the middle of the points.
(d) At engine 2.2 L, line gives roughly 180 - 195 g/km.

Question 18

(a) $V = L \times W \times H = 6 \times 8 \times 10 = 480 \text{ cm}^3$.
(b) A cuboid has three pairs of identical rectangular faces.
 $2(6 \times 8) + 2(6 \times 10) + 2(8 \times 10) = 2(48) + 2(60) + 2(80) = 96 + 120 + 160 = 376 \text{ cm}^2$.

Question 19

Angle per person = $360 / 60 = 6 \text{ deg}$.
Netflix: $20 \times 6 = 120 \text{ deg}$.
Disney+: $15 \times 6 = 90 \text{ deg}$.
Prime: $10 \times 6 = 60 \text{ deg}$.
Apple TV: $9 \times 6 = 54 \text{ deg}$.
Other: $6 \times 6 = 36 \text{ deg}$.
Check: $120 + 90 + 60 + 54 + 36 = 360 \text{ deg}$.
Use a protractor and label every sector.

Question 20

(a) Alex x , Ben $x+15$, Chris $2x$.

Total = $x + (x+15) + 2x = 4x + 15 = 155$. (Shown.)

(b) $4x = 140$, so $x = 35$.

(c) Chris = $2x = 2 \times 35 = £70$.

Check: Alex £35, Ben £50, Chris £70. Sum = £155.

Question 21

(a) Each month is 5% more, i.e. multiplier 1.05.

Feb = $120 \times 1.05 = £126.00$.

March = $126 \times 1.05 = £132.30$.

(b) Total = $120 + 126 + 132.30 = £378.30$.

Question 22

(a) $P(\text{yellow}) = 3 / 12 = 1/4$.

(b) $P(\text{not yellow}) = 1 - 1/4 = 3/4$ (or $9/12$).

(c) Expected frequency = probability $\times$ trials.

$P(\text{blue}) = 4 / 12 = 1/3$.

Expected number of blues = $(1/3) \times 60 = 20$.
